MIL-V-25513E (USAF)
13 September 1979
SUPERSEDING
MIL-V-25513D (USAF)
17 January 1974

MILITARY SPECIFICATION

# VALVE, CHECK, FOR 300 PSI LIQUID OXYGEN CONVERTER SYSTEMS, TYPE MH-1

This specification is approved for use by Code 99, Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers check valves for use in multiple installations of liquid oxygen converters having working pressures of 300 pounds per square inch gage (psig).

## 2. APPLICABLE DOCUMENTS

* 2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

| | |
|---|---|
| QQ-P-416 | - Plating, Cadmium (Electrodeposited) |
| PPP-T-60 | - Tape, Packaging, Waterproof |
| PPP-B-636 | - Box, Shipping, Fiberboard |

MILITARY

| | |
|---|---|
| MIL-P-116 | - Preservation-Packaging, Methods of |
| MIL-P-7105 | - Pipe Threads, Taper, Aeronautical National Form, Symbol ANPT, General Requirements for |

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: HQ AFLC CASO/LODS, Federal Center, Battle Creek, MI 49016, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC: 4820

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MIL-S-7742 - Screw Threads, Standard, Optimum Selected Series, General Specification for
MIL-A-8625 - Anodic Coatings, For Aluminum and Aluminum Alloys
MIL-O-27210 - Oxygen, Aviator's Breathing, Liquid and Gas
MIL-T-27730 - Tape, Antiseize, Tetrafluorethylene, With Dispenser

STANDARDS

MILITARY

MIL-STD-100 - Engineering Drawing Practices
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-130 - Identification Marking of U.S. Military Property
MIL-STD-143 - Specifications and Standards, Order of Precedence for the Selection of
MIL-STD-794 - Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-831 - Test Reports, Preparation of
MIL-STD-889 - Dissimilar Metals
MS33656 - Fitting End, Standard Dimensions for Flared Tube Connection and Gasket Seal

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

* 3.1 First article. When specified, the contractor shall furnish sample units for first article inspection and approval (see 4.2 and 6.2).

3.2 Selection of specifications and standards. Specifications and standards for necessary comodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.3 Materials.

3.3.1 Metals. [illegible] shall be of the corrosion-resistant type or suitably treated to resist corrosion due to fuels, salt spray or atmospheric condition likely to be met in storage or normal service.

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3.3.1.1 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.3.2 Nonmetallic materials. Any nonmetallic material that is easily deteriorated or otherwise easily affected adversely by continued service use with oxygen shall not be used.

3.4 Design. The valve shall conform to the design and dimensions shown on figure 1.

3.4.1 Flow direction and marking. Oxygen shall be permitted to flow in the direction indicated by an arrow, or arrows, marked or embossed on the valve exterior, and shall be checked in the opposite direction.

3.5 Construction. The valve shall operate as specified herein in any position in which it is mounted.

3.5.1 Valve body. The valve body shall be hexagonal and have flats to serve as wrench-gripping surfaces. The hexagonals or flats shall be 3/4 or 11/16 standard wrench flats.

3.6 Threads.

3.6.1 Screw threads. The threads for all machine screws shall conform to specification MIL-S-7742.

3.6.2 Pipe threads. Pipe threads shall conform to specification MIL-P-7105.

3.6.3 Lubricant. When necessary to prevent seizing of pipe threads, tape conforming to specification MIL-T-27730 shall be applied to male threads only. No other lubricant shall be permitted on the valve.

3.7 Finish.

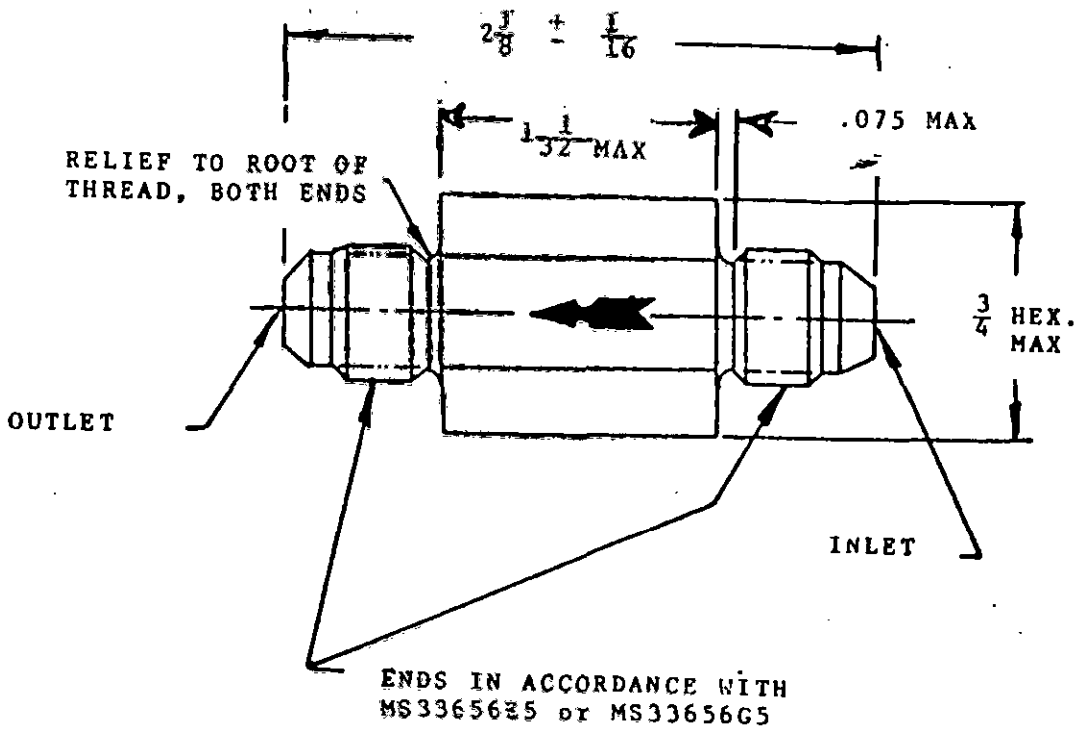
3.7.1 Aluminum and aluminum alloy parts shall be anodized in accordance with specification MIL-A-8625.

3.7.2 Ferrous materials. Ferrous materials shall be cadmium-plated in accordance with specification QQ-P-416, type II class 2.

3.8 Conditions. Valves shall be thoroughly dry and shall be free from oil or other foreign matter.



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NOTE 1 - OPPOSITE FITTING ENDS SHALL BE CONCENTRIC WITHIN .032 FULL INDICATOR READING

NOTE 2 - ALL DIMENSIONS IN INCHES-TOLERANCE: FRACTIONS ± 1/32

FIGURE 1. Converter check valve envelope.



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3.9 Performance.

3.9.1 Body leakage. The valve shall not leak when any pressure from zero to 450 psi is applied.

3.9.2 Flow check leakage. The checks in the direction in which the flow is checked shall be capable of withstanding a pressure of 10 to 450 psi without leakage in excess of one cubic centimeter per minute.

3.9.3 Valve leakage below opening pressure. The leakage through the valve when a pressure of 40 plus or minus 2 psi is applied to the inlet port shall not exceed 0.03 liter per minute.

3.9.4 Opening pressure. The valve shall open at 50 psi and shall allow a flow of at least 20 liters per minute at a maximum pressure of 60 psig.

3.9.5 Chattering. The valve shall not chatter when the inlet pressure is gradually increased (not less than 10 seconds) from zero to 100 psig.

3.9.6 Pressure drop. The pressure drop through the valve shall not exceed 20 psig in addition to the opening pressure when a pressure of 300 psi is applied at the inlet of the valve and a flow of 250 liters of free gas per minute is maintained through the valve.

3.9.7 High and low temperature operation. The valve shall be capable of operating without failure or damage to any of its parts at temperatures from minus 54°C to plus 127°C (minus 65°F to plus 260°F).

3.9.8 Torque. All pipe threads shall be capable of withstanding a maximum torque of 300 pound-inches without damage. All straight threads shall be capable of withstanding a maximum torque of 100 pound-inches without damage.

3.9.9 Vibration. The valves shall be capable of operating while being vibrated at a double amplitude (total displacement) of 0.018 to 0.020 inch at a frequency cycled from 300 to 3000 cycles per minute, for a period of six hours.

3.9.10 Low temperature exposure. The valve shall show no signs of failure of material or finish when exposed to temperature as low as minus 80 degrees Fahrenheit.



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3.9.11 Overpressure. The valve shall show no sign of structural failure or damage when a pressure of 600 psi is applied to the inlet or outlet of the valve for not less than one minute.

3.9.12 Excess flow. The valve shall show no signs of damage or malfunction of any parts when a pressure of 200 psi is applied to the inlet for not less than three minutes and the outlet is left open to the atmosphere.

3.9.13 Endurance check. The valve shall be cycled not less than 500 times while simulating actual operating conditions for the unit.

3.10 Weight. The weight of the valve shall not exceed 0.07 pound.

3.11 Marking. Direction of flow shall be marked as specified in 3.4.1.

* 3.12 Interchangeability. All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part number shall be governed by the drawing number requirements of MIL-STD-100.

3.13 Identification of product. The valves shall be marked in accordance with MIL-STD-130. The marking shall include the following:

VALVE, CHECK, 300 PSI, TYPE MH-1
MIL-V-25513D
Manufacturer's Identification .....
Contract or Order Nr .....

3.14 Workmanship. The valves shall be constructed in accordance with good commercial practice. They shall be free from any defect that might affect appearance or functionability. The valves shall be clean, dry and free of oil, grease or other foreign matter.

* 3.15 Test report. A test report in accordance with DI-T-3718A, paragraph 1.b, shall be prepared and submitted for approval (see 6.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless other specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as



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otherwise specified, the contractor may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Classification of tests. The inspection and testing of valves shall be classified as follows:

a. Acceptance tests

b. First article testing

4.3 Acceptance tests. Acceptance tests shall consist of:

a. Individual tests

b. Sampling tests

4.3.1 Individual tests. Each valve submitted for acceptance under contract shall be subjected to the following tests as described under 4.6:

a. Body leakage

b. Flow check leakage

c. Valve leakage below opening pressure

d. Opening pressure

e. Chattering

4.3.2 Sampling tests.

4.3.2.1 Lot. A lot shall consist of valves manufactured under essentially the same conditions and submitted for inspection at the same time.

4.3.2.1.1 Sampling plan A. Two valves shall be selected at random from each lot of 200 valves, or fraction thereof, on the order and subjected to the tests listed below and described under 4.6:

a. Examination of product

b. Pressure drop

c. High temperature operation



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d. Low temperature operation

e. Torque

4.3.2.1.2 Sampling plan B. Two valves shall be selected at random from each lot of 5,000, or fraction thereof, and subjected to the following tests, as described under 4.6:

a. Vibration

b. Low temperature exposure

c. Overpressure

d. Excess flow

e. Endurance check

4.3.2.1.3 Rejection and retest. When one or more items from a lot fail to meet the specification, acceptance of all items in the lot will be withheld until the extent and cause of failure are determined. After corrections have been made, all necessary tests shall be repeated.

4.3.2.1.4 Individual tests may continue. For production reasons, individual tests may be continued pending the investigation of a sampling test failure. Final acceptance of the entire lot shall not be made, however, until it is determined that the lot meets all the requirements of the specification.

4.3.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

* 4.4 First article testing.

4.4.1 First article testing sample testsed by the contractor. The contractor shall subject three valves to the first article tests specified in 4.4.4.

* 4.4.2 First article test report. After the contractor completes the first article tests, he shall prepare a first article test report in accordance with paragraphs 3.15 and 6.3.

* 4.4.3 First article test sample for the Government procuring activity. Along with the first article test report, the contractor shall submit one sample to the Government procuring activity specified in 6.3



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* 4.4.4 First article tests. First article tests shall consist of all tests described under 4.6.

4.5 Test conditions.

4.5.1 Atmospheric conditions. Unless otherwise specified, all tests required by this specification shall be conducted at an atmospheric pressure of 28 to 32 inches of mercury and at a temperature of 25 ± 10°C (77±18°F) and a relative humidity of not more than 80 percent. Where tests are made with atmospheric pressure or temperature different from the above values, proper allowance shall be made for the changes in instrument readings.

4.5.2 Vibration. Where a vibration stand is specified in the test procedure, it shall vibrate a cycled frequency between 300 and 3000 cycles per minute, at a double amplitude (total displacement) of 0.018 to 0.020 inch.

4.5.3 Test media. Oxygen conforming to specification MIL-O-27210, type I, shall be used as the high-pressure gas for the tests specified herein.

4.6 Test methods.

4.6.1 Body leakage. The inlet end of the valve shall be connected to a source of pressure and the outlet capped. Pressure shall be from zero to 45 psi. The valve shall be immersed in water or alcohol and examined for leakage through the assembly threads, while the pressure is increased slowly from zero to 450 psi. Evidence of leakage shall be cause for rejection.

4.6.2 Flow check leakage. The outlet end of the valve shall be connected to a pressure source which can be varied from zero to 450 psi, and the inlet end shall be connected to a suitable flow-measuring instrument. The valve shall be tested for leakage while the pressure is increased slowly from zero to 450 psi. Any leakage in excess of one cubic centimeter per minute shall be cause for rejection.

4.6.3 Valve leakage below opening pressure. A pressure of 40 psig, plus or minus 2 psig, shall be applied to the valve through the inlet port. The leakage through the valve shall not exceed 0.03 liter per minute.

4.6.4 Opening pressure. Pressure shall be gradually increased from zero psig to the opening pressure which shall be 50 psig, plus or minus 5 psig. At a maximum pressure of 60 psig, the valve shall allow a flow of at least 20 liters per minute. A flow of less than 20 liters per minute shall be cause for rejection.



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4.6.5 Chattering. The inlet pressure of the valve shall be gradually increased from zero to 100 psig in not less than 10 seconds with no evidence of chattering. There shall be no flow restrictions at the outlet during this test.

4.6.6 Examination of product. Samples selected shall be examined to determine conformance to this specification with respect to materials, workmanship, dimensions and markings.

4.6.7 Pressure drop. The pressure drop through the valves shall be measured when a pressure of 300 psig is applied at the inlet of the valve and a flow of 250 liters of free gas per minute is maintained through the valves. A pressure drop of more than 20 psig in addition to the opening pressure shall be cause for rejection.

4.6.8 High temperature operation. With a pressure of 450 psi applied to the outlet, the valves shall be subjected to a temperature of not less than 127°C (260°F) for a period of seven hours. During this test, the check valves shall be subjected to the flow check leakage test at least six times. Any evidence of failure or damage shall be cause for rejection.

4.6.9 Low temperature operation. With a pressure of 450 psi applied to the outlet, the valves shall be subjected to a temperature of minus 54 ± 1°C (minus 65 ± 2°F) for a period of seven hours. During this period, the samples shall be subjected to the flow check leakage test at least six times. Any evidence of failure or damage shall be cause for rejection.

4.6.10 Torque. The pipe threads shall be examined for evidence of failure after being subjected to a maximum of at least 300 pound-inches torque. The straight threads shall be examined for evidence of damage after being subjected to a maximum of 200 pound-inches torque. Any evidence of damage shall be cause for rejection.

4.6.11 Vibration. The valves shall be vibrated for a period of six hours while a pressure of 450 psi is applied to the inlet end with the outlet capped. During the vibration period, the samples shall be subjected to the flow check leakage test at least six times with the inlet pressure disconnected. Any evidence of damage or leakage shall be cause for rejection.

4.6.12 Low temperature exposure. The samples shall be exposed to a temperature of minus 65 ± 3°C (minus 85 ± 5°F) for a period of 48 hours. At the end of the 48 hours, the samples shall be returned to room temperature and subjected to the individual tests.



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4.6.13 Overpressure. A pressure of 600 psi shall be applied to the inlet for not less than one minute with the outlet capped. The pressure shall then be applied to the outlet with the inlet open. There shall be no sign of structural failure or damage to the valve components. Upon completion of the test, the valve shall be subjected to and meet the individual tests.

4.6.14 Excess flow. A pressure of 200 psi shall be applied to the inlet for not less than three minutes with the outlet open to the atmosphere. There shall be no blockage of flow or damage to the valve components. Upon completion of the test, the valve shall be subjected to and meet the individual tests.

4.6.15 Endurance check. The valve shall be located in a pressure line with a pressure of 300 psi applied in the direction of flow. An additional valve or a means of relieving the downstream side shall be depleted 500 times, of sufficient amounts to actuate the check valve. Chattering of the valve during the test shall be cause for rejection. Upon satisfactory completion of the test, the valve shall be subjected to and meet the individual tests specified herein.

4.6.16 Preservation and packaging. Unless otherwise specified in the purchase order or contract, the supplier is responsible for the performance of all inspections required as listed in table III, MIL-P-116, for the method of preservation being accomplished.

* 5. PACKAGING

* 5.1 Preservation-packaging. Preservation-packaging shall be level A or C as specified by the procuring activity.

* 5.1.1 Level A. Each valve shall be packaged in accordance with method 1A of MIL-P-116, without the application of a preservative compound. Cleaning shall be accomplished without immersion in solvent. The valve shall be placed within the water-vaporproof enclosure required to accomplish this method and sealed.

* 5.1.2 Level C. Each valve shall be packaged in a manner that will prevent physical damage or deterioration during transit from the supply source to the first receiving activity for immediate use. The package shall be capable of precluding contamination tht might impair the use or efficiency of the valve. The contractor's commercial packaging may be utilized when it conforms to criteria stated herein.

* 5.2 Packing. Unless otherwise specified by the procuring activity, the valve shall be provided packing which will afford



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adequate protection against damage during shipment, handling, intermediate storage and worldwide redistribution.

* 5.2.1 Level A. Valves packaged as specified in 5.1.1 shall be packed in fiberboard containers conforming to PPP-B-636, class weather-resistant, style RSC, grade V3s or V3c. As far as practical, exterior containers shall be uniform in shape and size, be of minimum cube and tare consistent with the protection required, and contain identical quantities. Closure and waterproofing shall be accomplished by sealing all seams, corners and manufacturer's joint with tape, two inches minimum width, conforming to PPP-T-60. Reinforcement requirements shall be applied in accordance with the appendix to PPP-B-636 using nonmetallic or tape banding.

* 5.2.2 Level B. Valves packaged as specified in 5.1.1 shall be packed in fiberboard containers conforming to style RSC, class domestic, minimum grade 200 of PPP-B-636. Containers shall be of uniform shape and size and contain identical quantities. The container closure shall be accomplished in accordance with method II, 30.3.1.2 of the appendix to PPP-B-636.

* 5.2.3 Level C. The packaged valves shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carrier rules and regulations.

* 5.3 Marking. All unit and shipping containers shall be marked in accordance with MIL-STD-129. In addition, the following precautionary marking shall appear on the unit and shipping container:

> Oxygen Components - All oil, grease, shop residue, or other contaminants have been removed.
>
> Do not open until ready for use.

6. NOTES

6.1 Intended use. The check valves covered by the specification are intended for use with multiple installation of 300 pounds per square inch gage liquid oxygen converters used in aircraft oxygen breathing systems.

* 6.2 Ordering data. Procurement documents should specify the following:



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a. First article requirements

b. Level of packaging and packing required

* 6.3 Data. Data generated by this document is not deliverable unless specified on the Contract Data Requirements List (DD Form 1423) referencing the appropriate data item description of the DOD Authorized Data List (TD-3). The data produced by this specification is DI-T-3718A, paragraph 1.b. (Reference paragraph 3.16)

* 6.4 Identification of changes. The margins of this specification are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

* 6.5 Reclaimed material. The use of reclaimed material shall be encouraged to the maximum extent possible.

Custodian:
Air Force - 99

Preparing Activity:
Air Force - 99

Review Activity:
DCSC-CS

Project Number:
4820-F412



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# SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-V-25513E(USAF) Valve, Check, for 300 psi Liquid Oxygen Converter Systems, Type MH-1

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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